2.1 Rates of Change and the Derivative

Name _____ Period ____ Date _____

AP Calculus 2.1 · Student sheet

I can…

- use the definition to find the derivative of a function at a number.
- write an equation of the tangent line to a graph.
- interpret the derivative as a velocity or a rate of change.

Vocabulary · fill in the missing word

Derivative of f at c — $f'(c)$ = the limit as $x \to c$ of $\frac{f(x) - f(c)}{x - c}$, if the limit _____

Tangent line — the line _____ $(c, f(c))$ with slope $f'(c)$

Instantaneous velocity — the _____ of the position function, $v(t_0) = s'(t_0)$

Instantaneous rate of change — the _____ $f'(c)$: how fast f changes with respect to x at c

Part 1 · Zoom in

Zoom in on $y = x^2$ at (1, 1).

Zoom level	Two points you read	Slope between them
Close		
Closer		
Very close		

Part 2 · Notes and examples (fill in each note, then work the example)

Derivative at c: $f'(c)$ = the limit as $x \to c$ of _____, if the limit exists.

Example 1

Use the definition to find $f'(2)$ for $f(x) = 2x^2 - 5x$.

Tangent line at $(c, f(c))$: $y - f(c) =$ _____$(x - c)$.

Example 2

Find an equation of the tangent line to $f(x) = x^3$ at (1, 1).

Example 3

Use the definition to find $f'(1)$ for $f(x) = \frac{1}{x}$, then find the tangent line at (1, 1).

Velocity: if s(t) is the position, the instantaneous velocity at t_0 is _____.

Example 4

An object's position is $s(t) = 10t^2$ centimeters after t seconds. Find its velocity at t = 3.

Rate of change: $f'(c)$ is the instantaneous rate of change of f with respect to x at x = _____.

Example 5

x: 0, 1, 2, 3, 4 and f(x): 0, 3, 12, 33, 72. Approximate $f'(2)$.

Part 3 · Partner practice

Partner 1

Use the definition to find $f'(4)$ for $f(x) = \sqrt{x}$.

Partner 2

(AP® Practice) The line x + y = 5 is tangent to the graph of y = f(x) at the point where x = 2. The values of f(2) and $f'(2)$ are

(A) f(2) = 2; $f'(2) = -1$ (B) f(2) = 3; $f'(2) = -1$

(C) f(2) = 2; $f'(2) = 1$ (D) f(2) = 3; $f'(2) = 1$